

Chapter 7: Alternating Current

Formula Sheet + Tips & Tricks

Formula Sheet

I. Mean, Peak and rms Values of AC

rms and mean value (sinusoidal AC)

$$I_{rms} = \frac{I_0}{\sqrt{2}}, \quad V_{rms} = \frac{V_0}{\sqrt{2}}, \quad I_{mean} = \frac{2I_0}{\pi}$$

I_0, V_0 = peak current/voltage; mean value is defined only over a half-cycle (it is zero over a full cycle).

rms value of a sine + cosine combination

$$I = A \sin \omega t + B \cos \omega t \Rightarrow I_0 = \sqrt{A^2 + B^2}, \quad I_{rms} = \frac{I_0}{\sqrt{2}}$$

A, B = amplitudes of the sine and cosine components (same frequency).

rms value with a DC offset

$$I = I_{dc} + I_{ac} \sin \omega t \Rightarrow I_{rms} = \sqrt{I_{dc}^2 + \frac{I_{ac}^2}{2}}$$

I_{dc} = steady (DC) part, I_{ac} = peak value of the varying part.

II. AC Circuit with either of R, L or C

Inductive and capacitive reactance

$$X_L = \omega L = 2\pi f L, \quad X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

X_L = inductive reactance, X_C = capacitive reactance, f = frequency, L = inductance, C = capacitance.

Phase relations for pure elements

Pure R : V, I in phase. Pure L :

V leads I by 90° . Pure C : I leads V by 90°

Governs the phase relation between voltage and current for a single circuit element.

III. AC Circuit containing R and L

Impedance and phase angle (R-L series)

$$Z = \sqrt{R^2 + X_L^2}, \quad \tan \phi = \frac{X_L}{R}, \quad I = \frac{V}{Z}$$

Z = impedance, ϕ = phase angle between V and I (current lags voltage).

IV. AC Circuit containing R and C

Impedance and phase angle (R-C series)

$$Z = \sqrt{R^2 + X_C^2}, \quad \tan \phi = \frac{X_C}{R}$$

Same impedance triangle as R - L , but with I leading V instead of lagging.

V. L-C-R Circuit

Impedance and phase angle (series L-C-R)

$$Z = \sqrt{R^2 + (X_L - X_C)^2}, \quad \tan \phi = \frac{X_L - X_C}{R}$$

If $X_L > X_C$ the circuit is inductive (current lags); if $X_C > X_L$ it is capacitive (current leads).

Resonance condition

$$X_L = X_C \Rightarrow f_0 = \frac{1}{2\pi\sqrt{LC}}, \quad Z_{min} = R$$

f_0 = resonant frequency; at resonance the circuit is purely resistive and current is maximum.

Quality factor

$$Q = \frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

Q = quality factor, a measure of the sharpness of resonance.

VI. L-C Circuits and Oscillations

Series L-C circuit

$$V = |V_L - V_C|, \quad Z = |X_L - X_C|$$

In a pure L - C series circuit, V_L and V_C are 180° out of phase, so they subtract, not add.

Minimum-impedance / maximum-current condition

$$\omega L = \frac{1}{\omega C}$$

Used to find the L or C needed to minimise impedance (maximum brightness/current).

VII. Power Consumed in AC Circuits

Average power and power factor

$$P = V_{rms} I_{rms} \cos \phi, \quad \cos \phi = \frac{R}{Z}$$

For a pure resistor $\phi = 0$, so $P = I_{rms}^2 R = V_{rms} I_{rms}$.

VIII. Transformer

Ideal transformer relation

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}, \quad \eta = \frac{P_{out}}{P_{in}} \times 100\%$$

N_p, N_s = primary/secondary turns; V_p, V_s, I_p, I_s = primary/secondary voltage and current; η = efficiency.

Tips, Tricks & Hacks

- rms value is what all AC meters read and what appears in power formulas: $I_{rms} = \frac{I_0}{\sqrt{2}}$, $V_{rms} = \frac{V_0}{\sqrt{2}}$ — never plug peak values directly into $P = VI$.
- Phase memory trick “CIVIL”: in a Capacitor, I leads V; in an Inductor, V leads I — instantly tells you the sign of the phase angle without re-deriving.
- Reactance formulas: $X_L = \omega L$ (grows with frequency), $X_C = \frac{1}{\omega C}$ (shrinks with frequency) — at very high frequency an inductor blocks and a capacitor passes; at $\omega \rightarrow 0$ it's the reverse.
- Resonance in LCR circuit occurs when $X_L = X_C$, i.e. $\omega_0 = \frac{1}{\sqrt{LC}}$ — at resonance impedance is purely resistive ($Z = R$) and current is maximum.
- Power factor $\cos \phi = \frac{R}{Z}$: a purely resistive circuit has $\cos \phi = 1$ (max power dissipated); a purely inductive or capacitive circuit has $\cos \phi = 0$ (zero average

power, the “wattless current”).

- Quality factor $Q = \frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$ tells you how sharp the resonance is — higher Q means a narrower resonance peak.
- Transformer: $\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$ — note current ratio is *inverted* compared to voltage/turns ratio; a step-up in voltage is always a step-down in current, and vice versa.
- An ideal transformer conserves power ($V_p I_p = V_s I_s$); real transformer losses (copper loss $I^2 R$, eddy current, hysteresis) are why efficiency $< 100\%$ — a common short-answer topic.